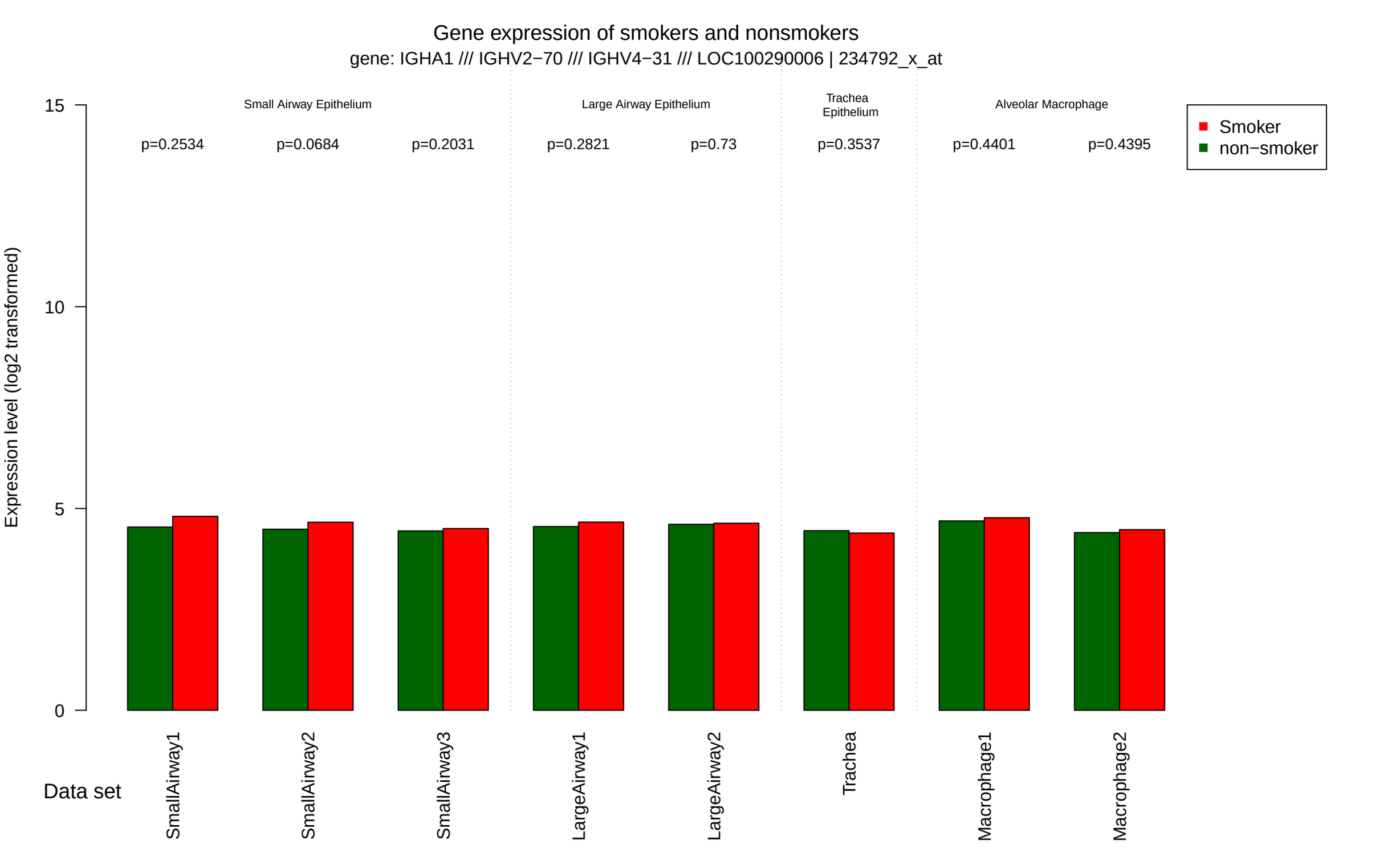
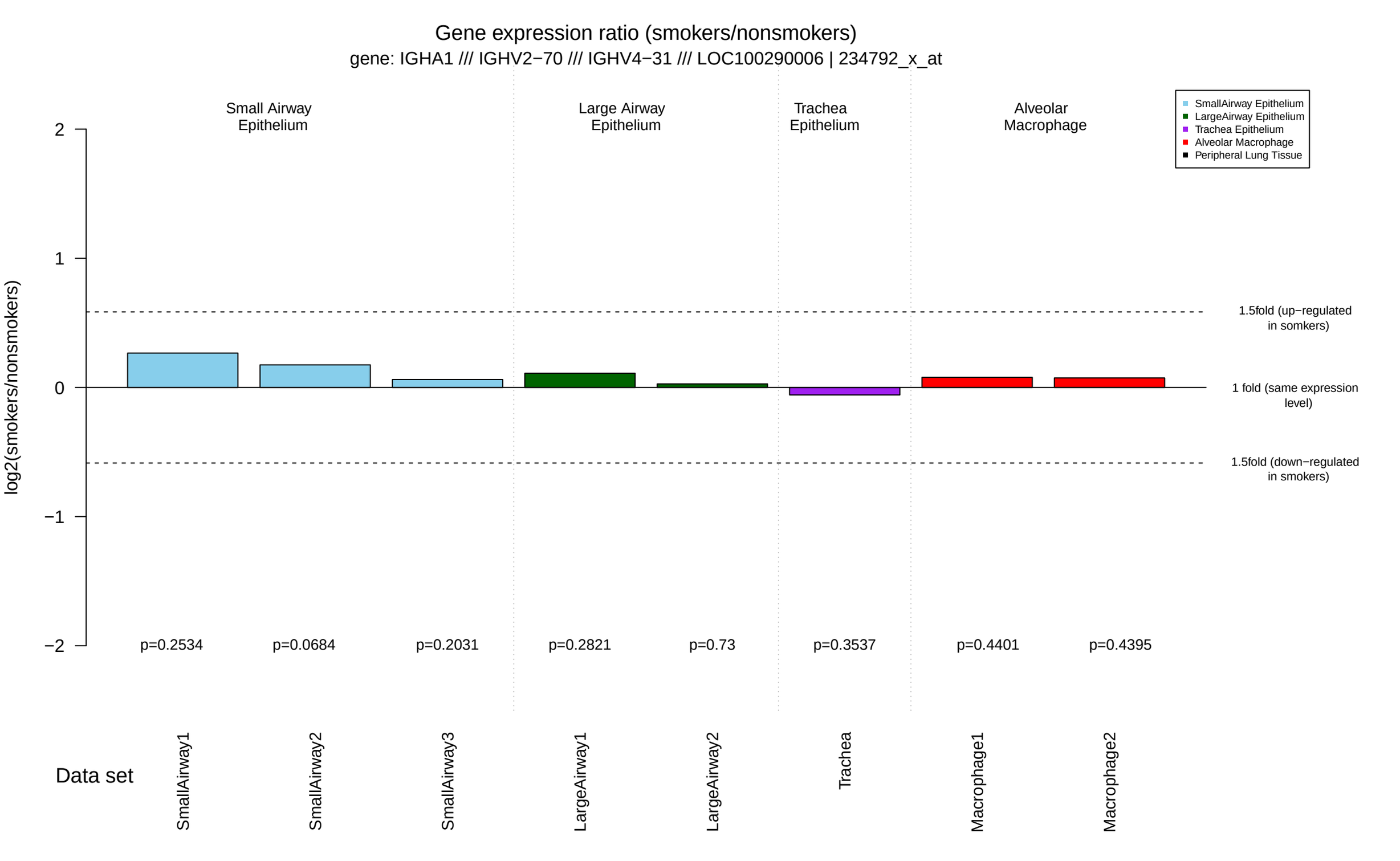
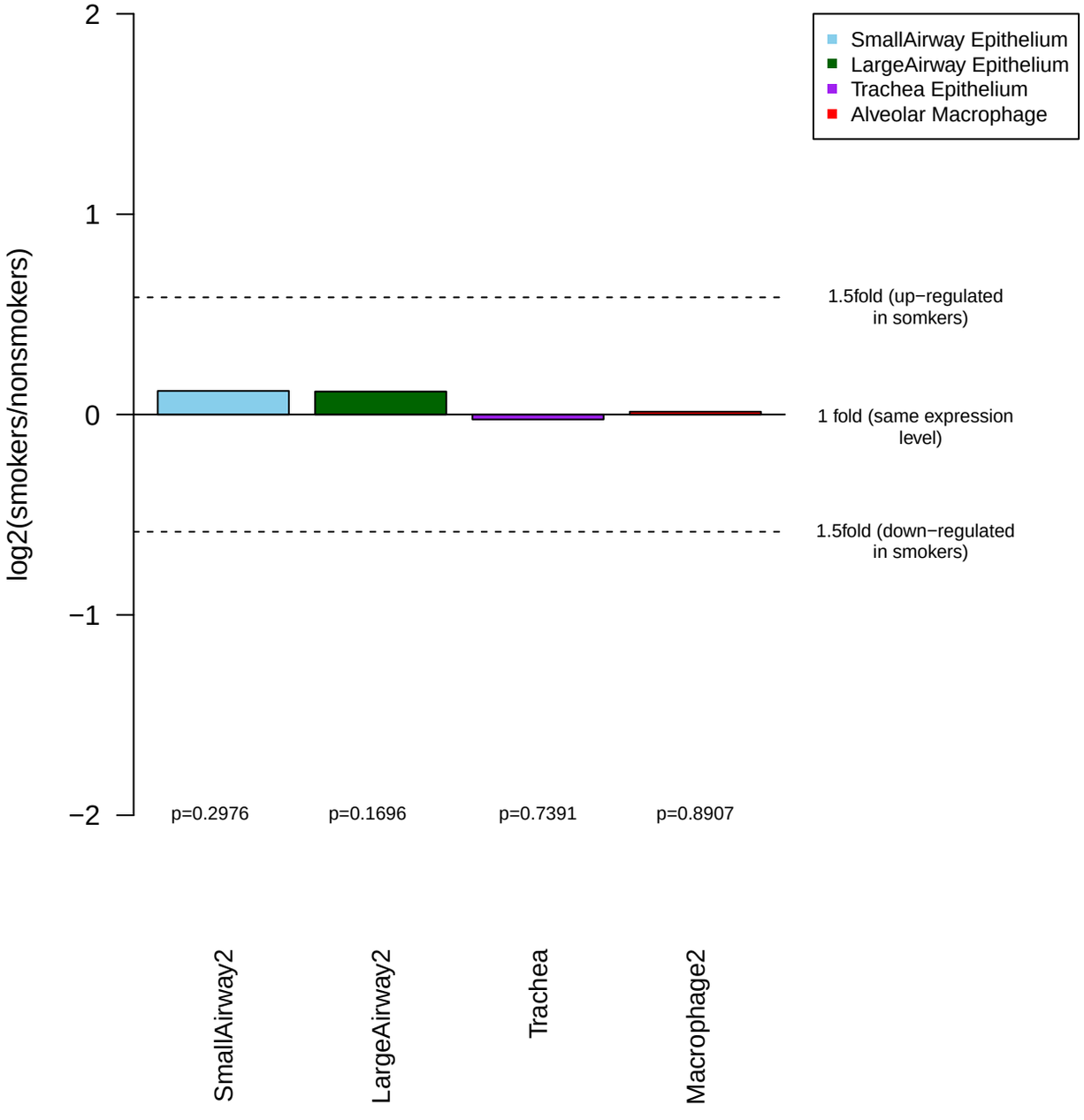

Gene expression of smokers and nonsmokers
gene: IGHA1 /// IGHV2−70 /// IGHV4−31 /// LOC100290006 | 234792_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2534
p=0.0684
p=0.2031
p=0.2821
p=0.73
p=0.3537
p=0.4401
p=0.4395
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2


Gene expression ratio (smokers/nonsmokers)
gene: IGHA1 /// IGHV2−70 /// IGHV4−31 /// LOC100290006 | 234792_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2534
p=0.0684
p=0.2031
p=0.2821
p=0.73
p=0.3537
p=0.4401
p=0.4395
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

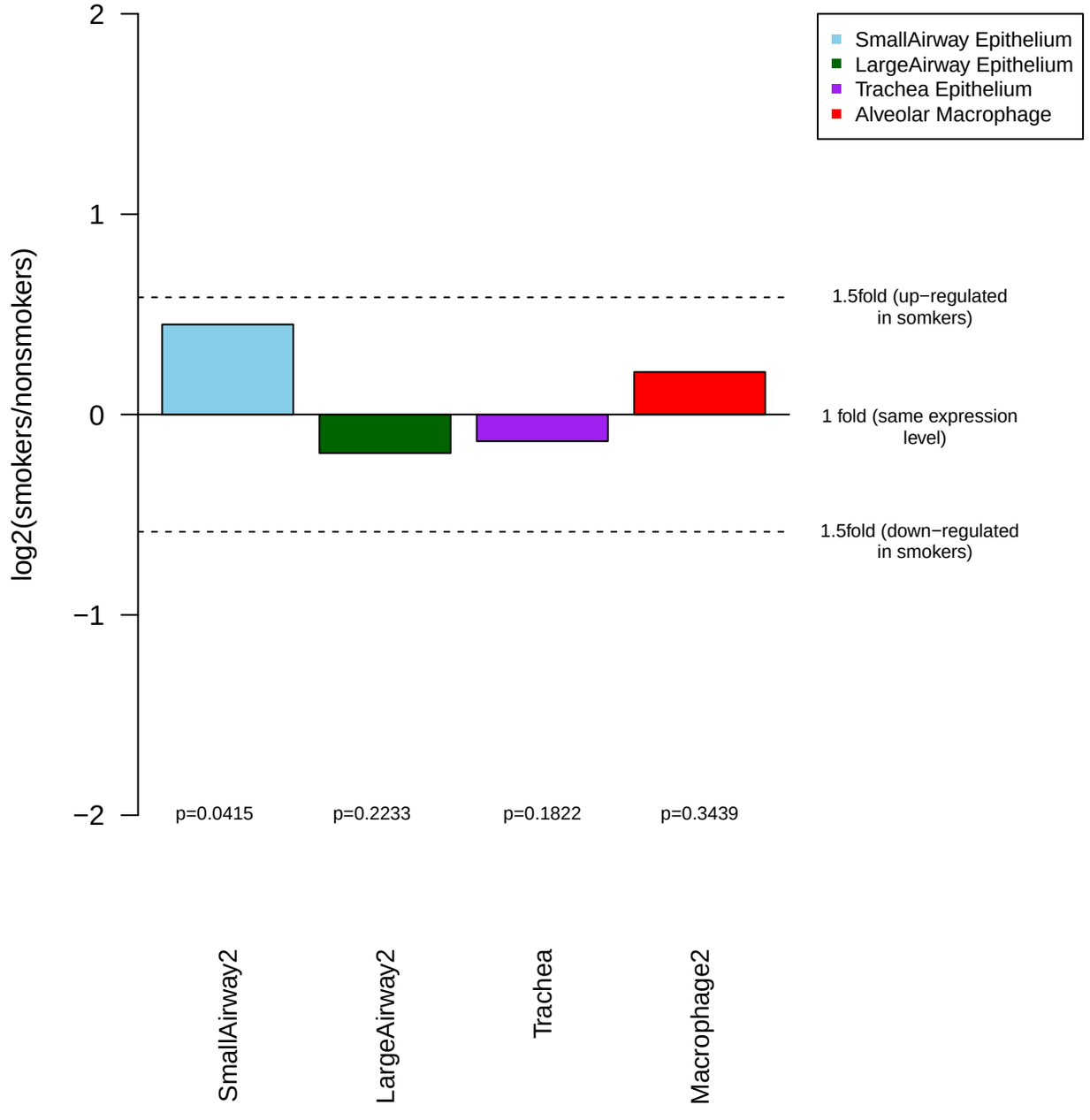
Gene expression ratio in Males

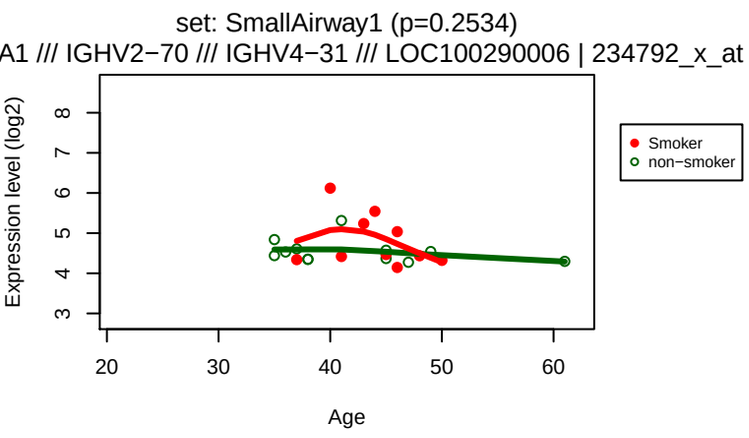
gene: IGHA1 /// IGHV2−70 /// IGHV4−31 /// LOC100290006 | 234792_x_at

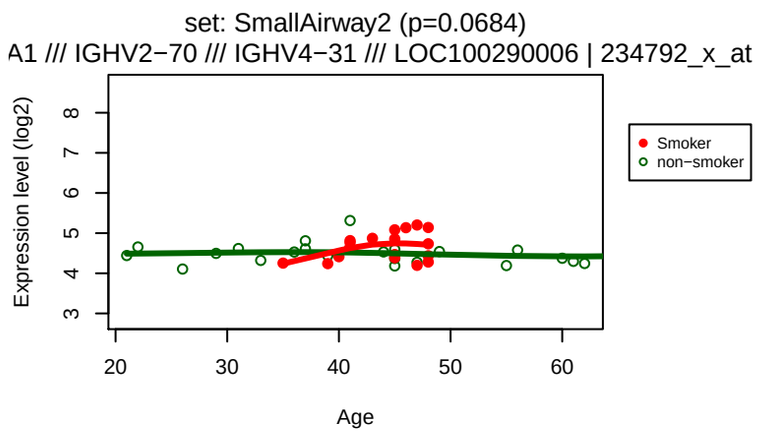


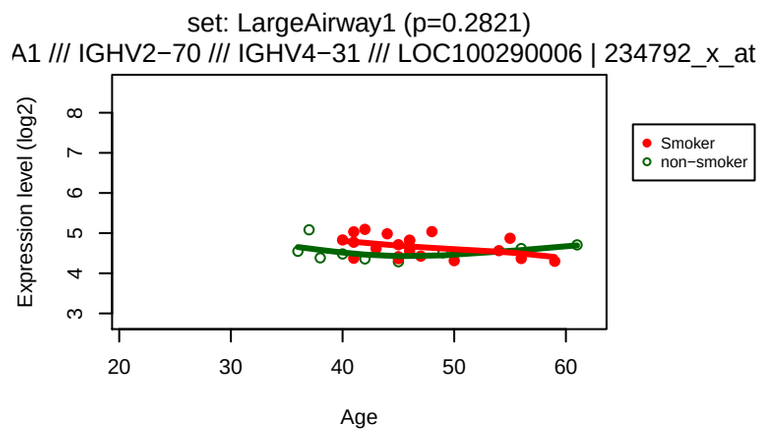
Gene expression ratio in Females

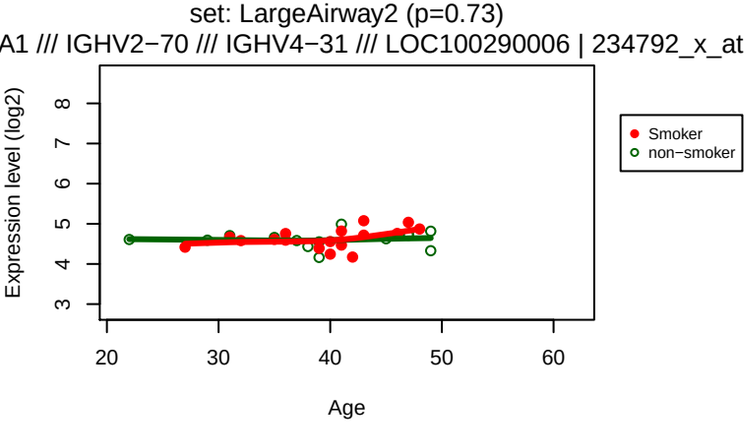
gene: IGHA1 /// IGHV2−70 /// IGHV4−31 /// LOC100290006 | 234792_x_at

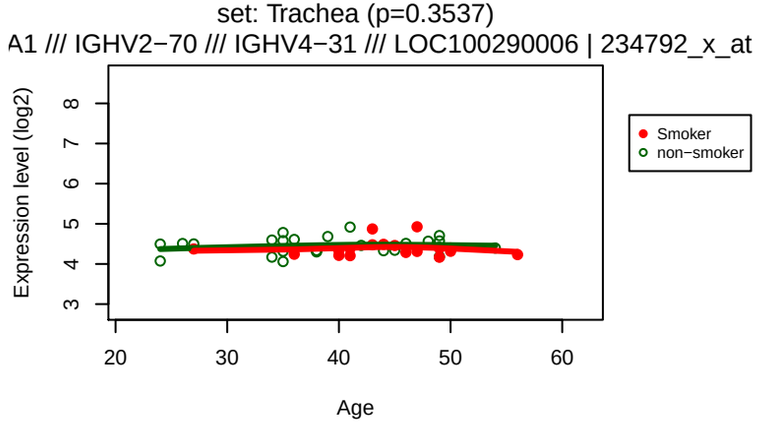


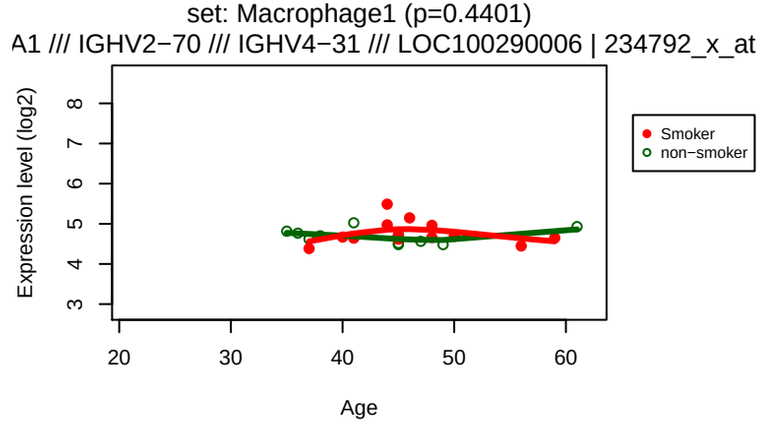

set: SmallAirway1 (p=0.2534)
A1 /// IGHV2−70 /// IGHV4−31 /// LOC100290006 | 234792_x_at
Expression level (log2)
Age
Smoker
non−smoker

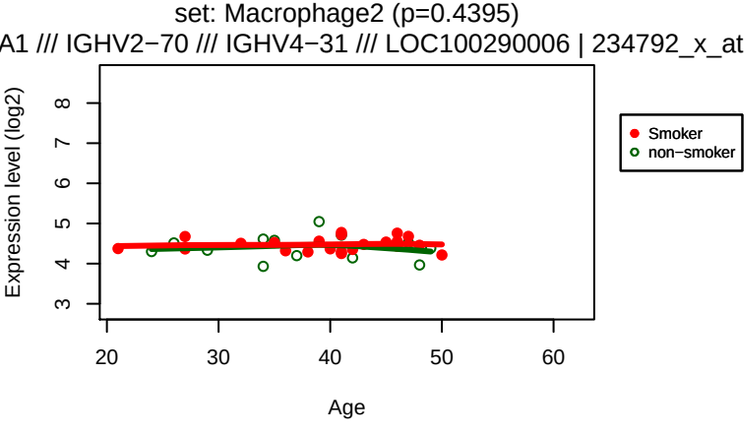

set: SmallAirway2 (p=0.0684)
A1 /// IGHV2−70 /// IGHV4−31 /// LOC100290006 | 234792_x_at
Expression level (log2)
Age
Smoker
non−smoker

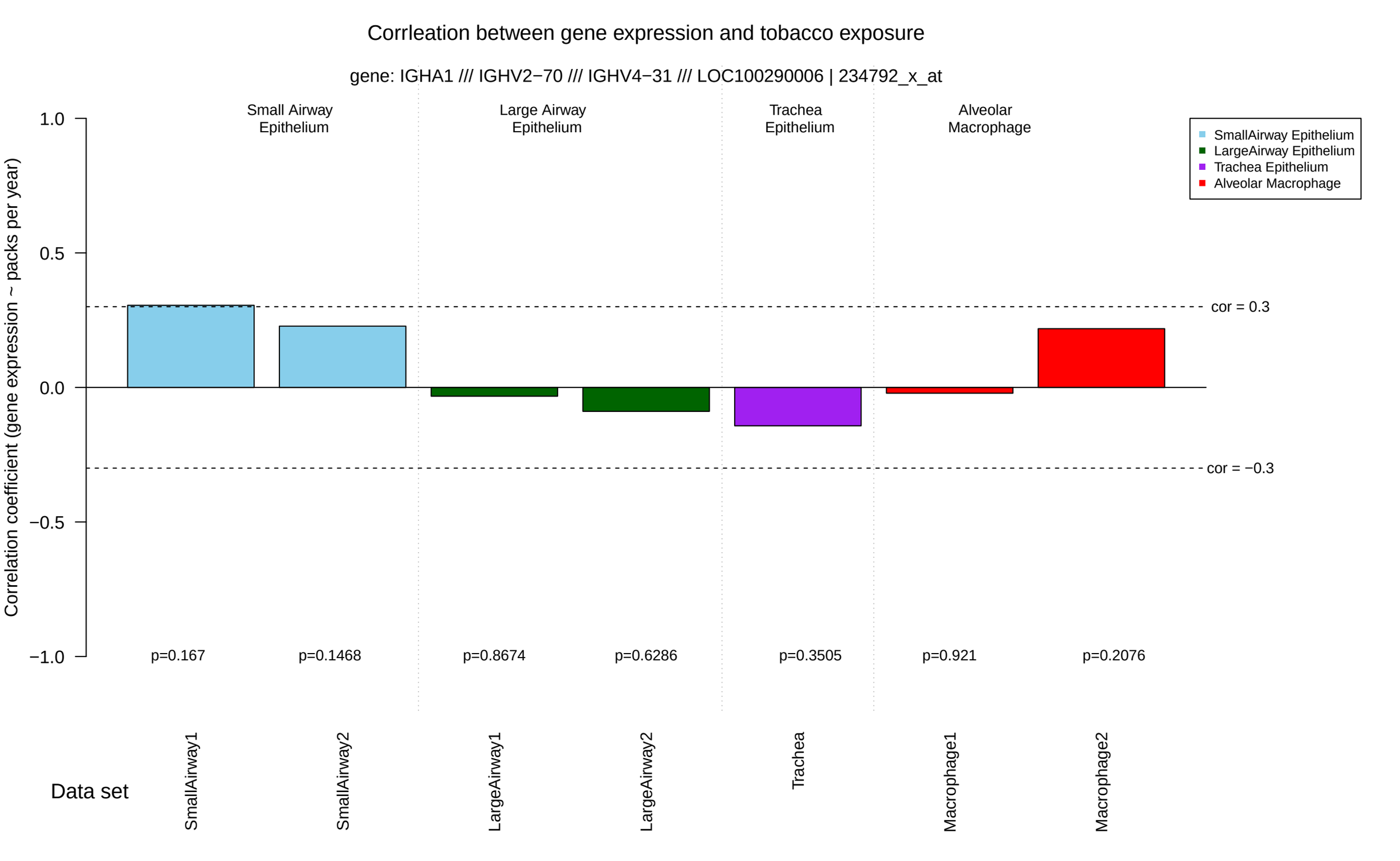

set: LargeAirway1 (p=0.2821)
A1 /// IGHV2−70 /// IGHV4−31 /// LOC100290006 | 234792_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.73)
A1 /// IGHV2−70 /// IGHV4−31 /// LOC100290006 | 234792_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3537)
A1 /// IGHV2−70 /// IGHV4−31 /// LOC100290006 | 234792_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4401)
A1 /// IGHV2−70 /// IGHV4−31 /// LOC100290006 | 234792_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4395)
A1 /// IGHV2−70 /// IGHV4−31 /// LOC100290006 | 234792_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGHA1 /// IGHV2−70 /// IGHV4−31 /// LOC100290006 | 234792_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.167
p=0.1468
p=0.8674
p=0.6286
p=0.3505
p=0.921
p=0.2076
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2